

1 Filter Specifications & Order Verification

1.1 Design Specifications

To achieve a **5th-order** Butterworth filter response, the system requirements must enforce a steep attenuation profile within a narrow transition band.

- **Passband Edge Frequency (f_p):** 1000 Hz
- **Maximum Passband Attenuation ($\alpha_{\max}$):** 3 dB (Cutoff frequency $f_c = f_p = 1000$ Hz)
- **Stopband Edge Frequency (f_s):** 2000 Hz
- **Minimum Stopband Attenuation ($\alpha_{\min}$):** 30 dB

1.2 Mathematical Determination of Filter Order (n)

The minimum required integer filter order is calculated using the standard analog Butterworth constraint equation:

$$n \geq \frac{\log_{10} \left(\frac{10^{0.1 \cdot \alpha_{\min}} - 1}{10^{0.1 \cdot \alpha_{\max}} - 1} \right)}{2 \cdot \log_{10} \left(\frac{f_s}{f_p} \right)}$$

Substituting the given design parameters into the equation:

$$n \geq \frac{\log_{10} \left(\frac{10^{3.0} - 1}{10^{0.3} - 1} \right)}{2 \cdot \log_{10} \left(\frac{2000}{1000} \right)}$$

$$\text{Numerator Component: } \log_{10} \left(\frac{1000 - 1}{1.9953 - 1} \right) = \log_{10} \left(\frac{999}{0.9953} \right) = \log_{10}(1003.717) \approx 3.0016$$

$$\text{Denominator Component: } 2 \cdot \log_{10}(2) = 2 \cdot 0.30103 = 0.60206$$

$$\text{Result: } n \geq \frac{3.0016}{0.60206} \approx 4.985$$

Rounding up to the nearest integer yields the structural constraint:

$$\mathbf{n = 5}$$

1.3 Poles Estimation

The poles for an odd 5th-order Butterworth system lie on the left half of the complex s -plane distributed along a unit circle. Spaced equally at angular increments of $\frac{\pi}{5}$ (36°), the angles are determined via:

$$\phi_k = \frac{\pi}{2} + \frac{(2k+1)\pi}{2n} \quad \text{for } k = 0,1,2,3,4$$

- s_0 ($k = 0$): $\phi_0 = 90^\circ + 18^\circ = 108^\circ \Rightarrow \mathbf{s_0 = -0.3090 + j0.9511}$
- s_1 ($k = 1$): $\phi_1 = 90^\circ + 54^\circ = 144^\circ \Rightarrow \mathbf{s_1 = -0.8090 + j0.5878}$
- s_2 ($k = 2$): $\phi_2 = 90^\circ + 90^\circ = 180^\circ \Rightarrow \mathbf{s_2 = -1.0000 + j0.0000}$ (Real Pole)
- s_3 ($k = 3$): $\phi_3 = 90^\circ + 126^\circ = 216^\circ \Rightarrow \mathbf{s_3 = -0.8090 - j0.5878}$
- s_4 ($k = 4$): $\phi_4 = 90^\circ + 162^\circ = 252^\circ \Rightarrow \mathbf{s_4 = -0.3090 - j0.9511}$

2 Transfer Function Expansion & Sallen-Key Topology

2.1 Expanding Conjugates to Determine Transfer Function

To form the algebraic denominator polynomial, complex conjugate pole pairs are grouped into real quadratic factors, while the lone real pole forms a single linear factor.

$$\text{Pair 1 } (s_0, s_4): (s + 0.3090 - j0.9511)(s + 0.3090 + j0.9511) = s^2 + 0.6180s + 1$$

$$\text{Pair 2 } (s_1, s_3): (s + 0.8090 - j0.5878)(s + 0.8090 + j0.5878) = s^2 + 1.6180s + 1$$

$$\text{Real Pole } (s_2): (s + 1)$$

Multiplying these expressions generates the complete normalized transfer function:

$$H(s) = \frac{1}{(s + 1)(s^2 + 0.6180s + 1)(s^2 + 1.6180s + 1)}$$

Applying frequency scaling ($s \rightarrow \frac{s}{\omega_c}$) where $\omega_c = 2\pi(1000) \approx 6283.185$ rad/s:

$$H(s) = \frac{\omega_c^5}{(s + \omega_c)(s^2 + 0.6180\omega_c s + \omega_c^2)(s^2 + 1.6180\omega_c s + \omega_c^2)}$$

2.2 Sallen-Key Circuit Implementation Diagram

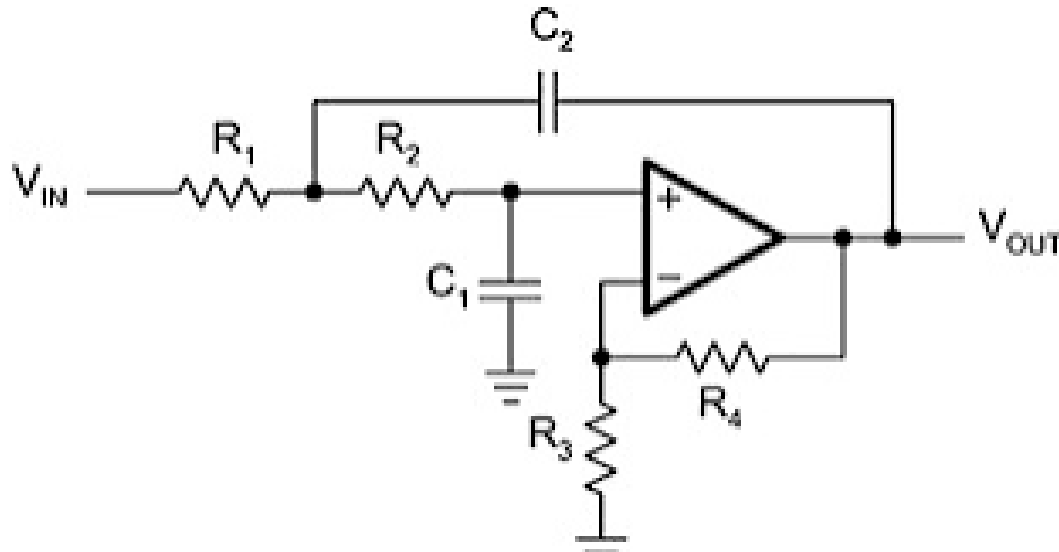


Figure 1: Sallen Key Topology

3 Component Values & Verification Framework

3.1 Numerical Value Computations

Component values are calculated based on a chosen system base resistance of $R = 10 \text{ k}\Omega$ and angular cutoff frequency $\omega_c \approx 6283.185 \text{ rad/s}$. The baseline constant factor is $\omega_c R = 62831.85$.

Stage 1 (1st-Order Real Pole Filter):

The mathematical cutoff constraint is $C_{1a} = \frac{1}{\omega_c R}$:

$$C_{1a} = \frac{1}{62831.85} \approx 15.92 \text{ nF} \Rightarrow \mathbf{16 \text{ nF}}$$

Stage 2 (2nd-Order Sallen-Key, $Q_1 = \frac{1}{0.6180} \approx \mathbf{1.6180}$):

$$C_{1b} = \frac{2Q_1}{\omega_c R} = \frac{2(1.6180)}{62831.85} \approx 51.50 \text{ nF} \Rightarrow \mathbf{51 \text{ nF}}$$

$$C_{2b} = \frac{1}{2Q_1\omega_c R} = \frac{1}{2(1.6180)(62831.85)} \approx 4.91 \text{ nF} \Rightarrow \mathbf{4.7 \text{ nF}}$$

Stage 3 (2nd-Order Sallen-Key, $Q_2 = \frac{1}{1.6180} \approx \mathbf{0.6180}$):

$$C_{1c} = \frac{2Q_2}{\omega_c R} = \frac{2(0.6180)}{62831.85} \approx 19.67 \text{ nF} \Rightarrow \mathbf{20 \text{ nF}}$$

$$C_{2c} = \frac{1}{2Q_2\omega_c R} = \frac{1}{2(0.6180)(62831.85)} \approx 12.87 \text{ nF} \Rightarrow \mathbf{13 \text{ nF}}$$

Hardware Implementation Parameters Summary Table

Component Designation	Ideal Analytical Value	Selected Production Component (E24)
All System Resistors ($R_1 \dots R_5$)	10.00 k Ω	10 k Ω
Stage 1 Capacitor (C_{1a})	15.92 nF	16 nF
Stage 2 Feedback Capacitor (C_{1b})	51.50 nF	51 nF
Stage 2 Shunt Capacitor (C_{2b})	4.91 nF	4.7 nF
Stage 3 Feedback Capacitor (C_{1c})	19.67 nF	20 nF
Stage 3 Shunt Capacitor (C_{2c})	12.87 nF	13 nF